Mini Project: Weather Data Dashboard

Weather stations on campus record temperature, humidity and wind speed every ten minutes. Your team will turn these readings into a simple dashboard.

What the dashboard has to do

The dashboard must read the CSV files that the stations produce. It must show a line chart of temperature for any chosen day. Users should be able to compare two stations on the same chart. The dashboard must flag readings that are missing for more than one hour.

Technical notes

Use Python with pandas and matplotlib or Plotly. The dashboard must open in under five seconds for one month of data.

Hand-in

Hand in a Jupyter notebook or a small web app, plus a one-page summary of your findings.

Criterion	Marks
Reading and cleaning the data	8
Charts	8
Missing-data flags	4

Total: 20 marks

1 The station files use UTC timestamps. Footnotes give context only.